

SIMATS ENGINEERING
ASSESSMENT TOOL 2: Case Study

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1516

COURSE OUTCOME COVERED

CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications. (BL3)*

Dave's Taxonomy: Precision (Level 3)
Question Count: 3

Total Marks: 30

Code of Conduct

I_Y SRAVAN KUMAR (192472289)_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____ SRAVAN _____

Assessment Tasks

Case Study

1. Case Study 1: Smart Retail Inventory Management System

A retail chain implements a cloud-based inventory system connected to store devices. Clients (POS systems and dashboards) communicate via APIs to centralized cloud services. Cloud storage stores product data, transaction logs, and analytics datasets. The system uses platform services for analytics and demand forecasting. Secure access is maintained through VPNs, firewalls, and API gateways. This solution integrates infrastructure, services, and web applications for efficient operations.

2. Case Study 2: Cloud-Based Document Collaboration System

An enterprise deploys a document collaboration tool similar to Google Docs. Users access the platform via browsers, enabling real-time editing and sharing. Documents are stored in distributed cloud storage with version control mechanisms. Web APIs enable synchronization across multiple clients and devices. Network infrastructure ensures low latency and high availability globally. Security includes encryption, identity management, and access permissions for collaboration.

3. Case Study 3: Online Food Ordering Platform

A startup builds a cloud-based food ordering system accessible via web browsers and mobile clients. Frontend clients interact with backend services through RESTful Web APIs hosted on a cloud platform. Cloud storage is used to store menu images, invoices, and customer data securely. The infrastructure uses virtual machines and managed Kubernetes for scalability and reliability. Security is enforced

through authentication tokens, HTTPS, and role-based access control. The system demonstrates integration of clients, APIs, storage, and platform services in real time.

Case Study Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Case	25%	Clearly identifies all technical and business requirements	Identifies most requirements	Basic understanding only	Poor understanding of case
Application of Concepts	25%	Applies suitable cloud concepts and tools effectively	Good application with minor issues	Partial application	Weak or incorrect application
Analysis	20%	Strong analysis with clear trade-off reasoning	Reasonable analysis	Limited analysis	Minimal analysis
Solution Design	20%	Solution is feasible, practical, and well justified	Reasonable solution with minor gaps	Basic solution	Unclear solution
Clarity	10%	Well-structured and easy to follow	Mostly clear	Somewhat unclear	Poorly organized

CASE STUDY 1: SMART RETAIL INVENTORY MANAGEMENT SYSTEM

1. Introduction

A retail chain may have multiple stores located in different places. Managing product inventory, sales transactions, and stock availability manually can be difficult. To solve this problem, the company implements a **cloud-based inventory management system**.

The system connects store devices such as **Point-of-Sale (POS) systems and dashboards** with centralized cloud services. The cloud provides storage, processing, analytics, and forecasting capabilities.

2. Problem Identification

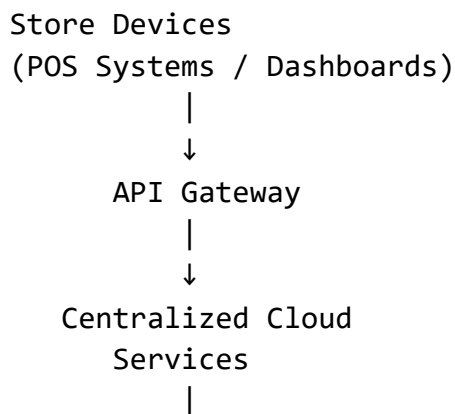
The retail company needs to:

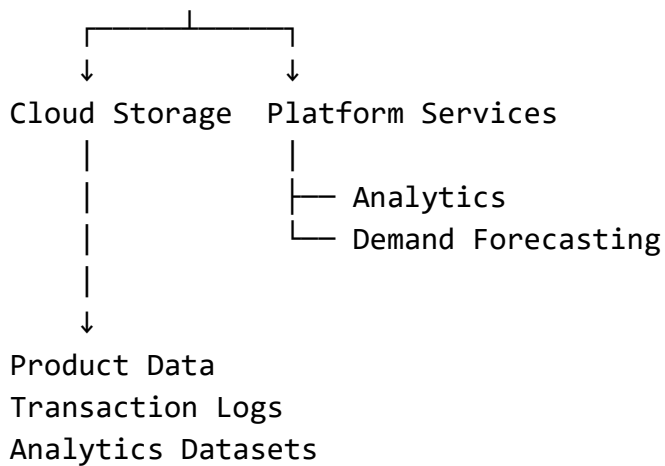
- Maintain product information centrally.
- Track transactions from different stores.
- Monitor inventory levels.
- Store large amounts of transaction data.
- Analyze sales information.
- Forecast future product demand.
- Provide secure access to cloud services.

A cloud-based solution can address these requirements by providing centralized services that can be accessed by different stores.

3. Cloud Architecture

The major components of the system are:





4. Components Used

A. Client Devices

POS systems and dashboards act as clients.

POS systems are used to record sales transactions, while dashboards can provide information about inventory and business operations.

B. APIs

APIs provide communication between the store devices and centralized cloud services.

For example:

POS System → API → Cloud Service

The API allows the application to send and receive inventory and transaction information.

C. Cloud Storage

Cloud storage stores:

- Product information
- Transaction logs
- Analytics datasets

Centralized storage allows information from different stores to be maintained in the cloud.

D. Platform Services

Platform services are used for:

- Data analytics

- Demand forecasting

Analytics can help the company understand sales patterns, while demand forecasting can help estimate future product requirements.

E. Security Components

The system uses:

- VPNs
- Firewalls
- API gateways

These mechanisms help protect communication, infrastructure, and API access.

5. Working of the System

1. A customer purchases a product at a retail store.
2. The POS system records the transaction.
3. The transaction information is sent to cloud services through an API.
4. Cloud services process the information.
5. Product and transaction information is stored in cloud storage.
6. Analytics services process the stored data.
7. Demand forecasting services analyze historical information.
8. Store dashboards can access relevant inventory information.
9. VPNs, firewalls, and API gateways provide secure access.

6. Security

Security is an important part of the system.

VPN: Provides secure connectivity between store environments and cloud services.

Firewall: Helps control and filter network traffic.

API Gateway: Controls API requests and provides a secure entry point to cloud services.

7. Benefits

The system provides:

1. Centralized inventory management.
2. Cloud-based data storage.
3. Integration between multiple stores.
4. Real-time communication through APIs.

5. Analytics capabilities.
6. Demand forecasting.
7. Secure access.
8. Better operational efficiency.

8. Conclusion

The Smart Retail Inventory Management System demonstrates how **cloud infrastructure, cloud storage, APIs, platform services, analytics, and security mechanisms** can be integrated to manage retail inventory efficiently.

CASE STUDY 2: CLOUD-BASED DOCUMENT COLLABORATION SYSTEM

1. Introduction

An enterprise requires employees to create, edit, store, and share documents. A cloud-based document collaboration system similar to **Google Docs** can provide these facilities through web browsers.

Users can access documents from different devices while the documents are stored in distributed cloud storage.

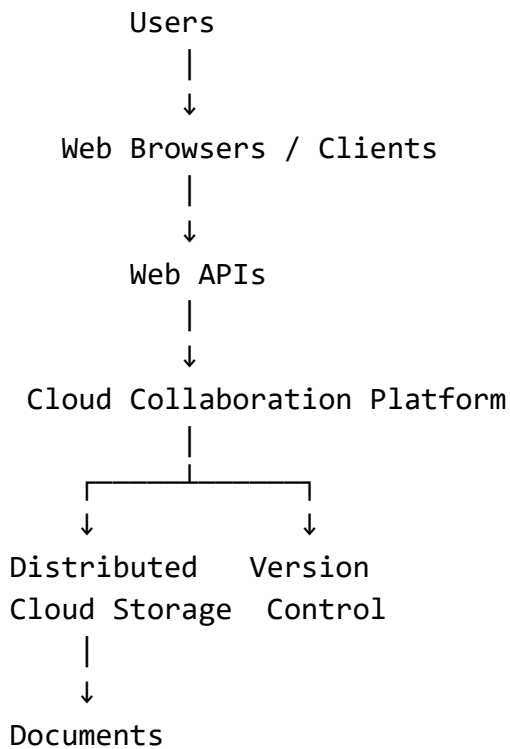
2. Problem Identification

A traditional document system may make collaboration difficult because users may need to exchange files manually.

The cloud-based system provides:

- Online document access.
- Real-time editing.
- Document sharing.
- Version control.
- Synchronization across devices.
- Secure access.

3. Cloud Architecture



4. Components Used

A. Web Browsers

Users access the collaboration platform through web browsers.

This means users do not necessarily need a separate desktop application to access their documents.

B. Cloud Platform

The cloud platform hosts the document collaboration application and provides the services required for document access and collaboration.

C. Distributed Cloud Storage

Documents are stored using distributed cloud storage.

This allows documents to be available through the cloud and supports access from different locations.

D. Version Control

Version control maintains different versions of documents.

For example:

Document
↓
Version 1
↓
Version 2
↓
Version 3

This allows previous versions to be maintained.

E. Web APIs

Web APIs synchronize information between multiple clients and devices.

For example:

Laptop
↕
Web API
↕
Cloud Document
↕
Mobile Device

5. Working of the System

1. A user opens the document collaboration platform using a browser.
2. The user logs into the system.
3. The user opens an existing document or creates a new document.
4. The document is stored in cloud storage.
5. Users can share the document with other users.
6. Multiple users can edit the document in real time.
7. Changes are synchronized using web APIs.
8. Version control maintains document versions.
9. Identity management verifies users.
10. Access permissions determine who can view or edit documents.

6. Security

The system uses several security mechanisms.

Encryption

Encryption protects documents and communication from unauthorized access.

Identity Management

Identity management identifies and authenticates users.

Access Permissions

Access permissions determine whether a user can:

- View a document
- Edit a document
- Share a document

7. Availability and Performance

The system uses network infrastructure designed for:

- Low latency
- High availability
- Global access

This allows users from different locations to access and collaborate on documents.

8. Benefits

The system provides:

1. Real-time collaboration.
2. Easy document sharing.
3. Cloud-based storage.
4. Version management.
5. Synchronization across devices.
6. Global accessibility.
7. Secure document management.
8. Reduced dependency on local storage.

9. Conclusion

The Cloud-Based Document Collaboration System demonstrates how **cloud storage, web APIs, distributed infrastructure, version control, and security services** can be integrated to provide real-time and secure document collaboration.

CASE STUDY 3: ONLINE FOOD ORDERING PLATFORM

1. Introduction

A startup develops a cloud-based food ordering platform that allows customers to order food through **web browsers and mobile applications**.

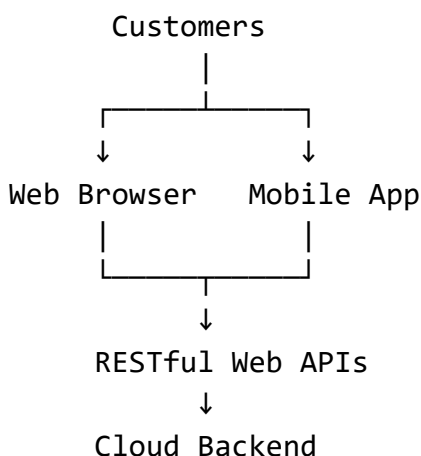
The system uses RESTful Web APIs to connect frontend clients with backend cloud services. Cloud storage maintains application data, while virtual machines and managed Kubernetes provide computing resources and scalability.

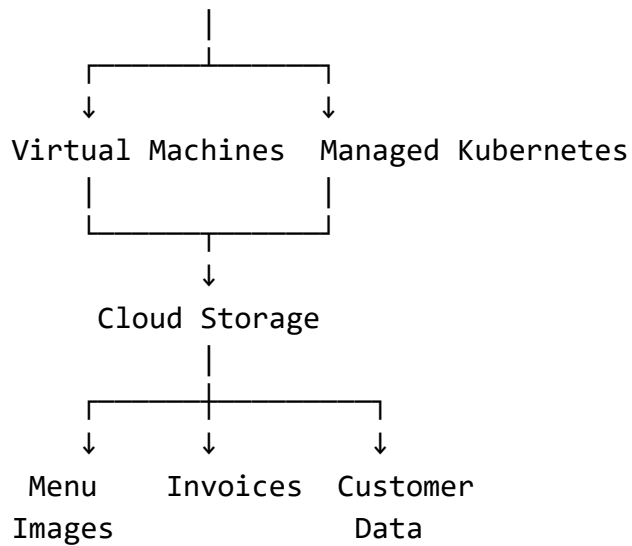
2. Problem Identification

The food ordering platform needs to:

- Support web users.
- Support mobile users.
- Process customer orders.
- Store customer information.
- Store menu images.
- Generate and store invoices.
- Handle multiple users simultaneously.
- Provide secure access.
- Scale when the number of users increases.

3. Cloud Architecture





4. Components Used

A. Web Browser

Customers can access the food ordering system through a web browser.

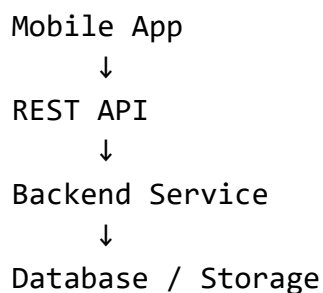
B. Mobile Clients

Mobile applications allow customers to browse menus and place orders from smartphones.

C. RESTful Web APIs

RESTful APIs provide communication between the frontend applications and backend services.

For example:



D. Cloud Platform

Backend services are hosted on a cloud platform.

The cloud provides computing resources required to process customer requests and orders.

E. Cloud Storage

Cloud storage is used to store:

- Menu images
- Invoices
- Customer data

F. Virtual Machines

Virtual machines provide computing infrastructure for running application services.

G. Managed Kubernetes

Managed Kubernetes is used to manage application containers and support scalability and reliability.

When demand increases, additional application resources can be provided to handle the workload.

5. Working of the System

1. A customer opens the website or mobile application.
2. The customer views the available food menu.
3. Menu information and images are retrieved from cloud services.
4. The customer selects food items.
5. The application sends the order through a RESTful API.
6. Backend cloud services process the order.
7. Customer and order information is stored securely.
8. An invoice is generated and stored in cloud storage.
9. Kubernetes manages the application services.
10. The system can scale when many customers access the platform simultaneously.

6. Security

The system uses multiple security mechanisms.

Authentication Tokens

Authentication tokens are used to verify users when they access the system.

HTTPS

HTTPS protects communication between customers and cloud services.

For example:

Customer



HTTPS



Cloud Application

This helps protect data during transmission.

Role-Based Access Control

RBAC controls what different users are allowed to access.

For example:

User	Access
Customer	Place and view orders
Restaurant Staff	Manage food orders
Administrator	Manage system

7. Scalability

The system uses **managed Kubernetes** to support scalability.

When the number of customers increases:

Low Traffic



Few Application Instances

High Traffic



More Application Instances

This helps the application handle increased workloads.

8. Reliability

Virtual machines and managed Kubernetes provide infrastructure for running the application reliably.

If demand changes, the cloud infrastructure can adjust the resources required by the application.

9. Benefits

The system provides:

1. Web and mobile accessibility.
2. Real-time API communication.
3. Cloud-based data storage.
4. Scalable application infrastructure.
5. Secure customer communication.
6. Authentication and authorization.
7. Support for large numbers of users.
8. Centralized management of application services.
9. Reliable food-order processing.

10. Conclusion

The Online Food Ordering Platform demonstrates the integration of **web and mobile clients, RESTful Web APIs, cloud storage, virtual machines, managed Kubernetes, authentication tokens, HTTPS, and role-based access control.**

The combination of these cloud technologies provides a **scalable, reliable, and secure platform** for online food ordering.